THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Covering iron with a layer of zinc is called: 1

(a) rancidity **(b)** electrolysis **(c)** galvanisation **(d)** distillation

2. The formula of rust is: 1

(a) $Fe(OH)_2$ **(b)** $FeSO_4$ **(c)** $Fe_2O_3 \cdot x\,H_2O$ **(d)** FeO

3. The reaction of quick lime with water is a: 1

(a) double displacement reaction **(b)** decomposition reaction
(c) displacement reaction **(d)** combination reaction

4. The colour of silver chloride is: 1

(a) blue **(b)** yellow **(c)** black **(d)** white

5. The gas evolved when zinc reacts with dilute sulphuric acid is: 1

(a) hydrogen **(b)** oxygen **(c)** nitrogen **(d)** carbon dioxide

6. Decomposition by heat is called: 1

(a) thermal decomposition **(b)** electrolysis
(c) neutralisation **(d)** photolysis

7. Which gas is used to flush packets of chips? 1

(a) oxygen **(b)** nitrogen **(c)** chlorine **(d)** hydrogen

8. A reaction is represented by an arrow (->) which means: 1

(a) 'is greater than' **(b)** 'gives' or 'yields'
(c) 'is equal to' **(d)** 'is dissolved in'

9. Decomposition by light is called: 1

(a) electrolytic decomposition **(b)** thermal decomposition
(c) photolytic decomposition **(d)** combination

10. The black coating formed on heated copper is: 1

(a) CuO **(b)** Cu_2S **(c)** $CuSO_4$ **(d)** $CuCl_2$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name two methods to prevent rusting of iron. 2

12. Define a combination reaction and give an example. 2

13. Define oxidation and reduction in terms of oxygen. 2

14. Define an exothermic reaction. Give one example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Balance the equation $\mathrm{Zn + HCl \longrightarrow ZnCl_2 + H_2}$. 3

16. Name the type of reaction: $\mathrm{2\,H_2O_2 \longrightarrow 2\,H_2O + O_2}$. Give a reason. 3

17. State any three observations that show a chemical reaction has taken place. 3

OR

Balance the equation $\mathrm{CH_4 + O_2 \longrightarrow CO_2 + H_2O}$. 3

18. Name the type of reaction: $\mathrm{NH_3 + HCl \longrightarrow NH_4Cl}$. Give a reason. 3

OR

Write the full balanced equation for the electrolysis of water and name the product at each electrode. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Differentiate between combination and decomposition reactions with two examples of each. 5

OR

Silver chloride is exposed to sunlight. State the observation, write the equation, type, and one use of the reaction. 5

20. Write one example each of combination, decomposition, displacement, double displacement and redox reaction. 5

OR

Define oxidation and reduction. For $\mathrm{CuO + H_2 \longrightarrow Cu + H_2O}$, name the substance oxidised, the substance reduced and the reducing agent. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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